Images

February 25, 2026

The mass of the Z boson was found to be 91.143 ± 0.009 ± 0.003 GeV where the first error is statistical and the second systematic
The width of the Z boson was found to be 2.026 ± 0.025 ± 0.000 GeV where the first error is statistical and the second systematic
The correlation coefficient was 0.065

	Mass	Width
Calibration error	-0.00299	0.00036
Smearing error	-0.00001	-0.00012
Υ(1S) mass error	0.0001	0.0000

Table 1: Z Systematic error on mass and width

	Mass	Width
Values	91.14260	2.02597
Statistical error	0.00872	0.02503

Table 2: Z Mass and width results with statistical error

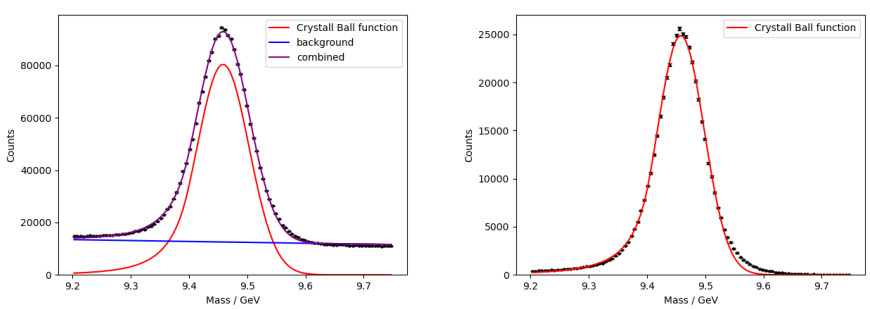

Figure 1:

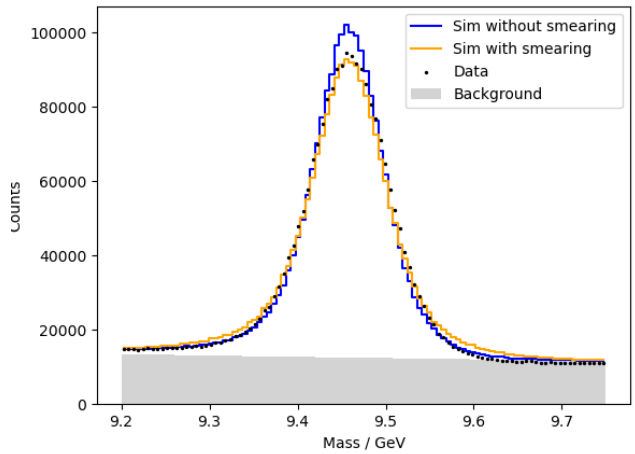

Figure 2:

χ^2	ndf
405.70	48

Table 3: minimum χ^2 and number of degrees of freedom

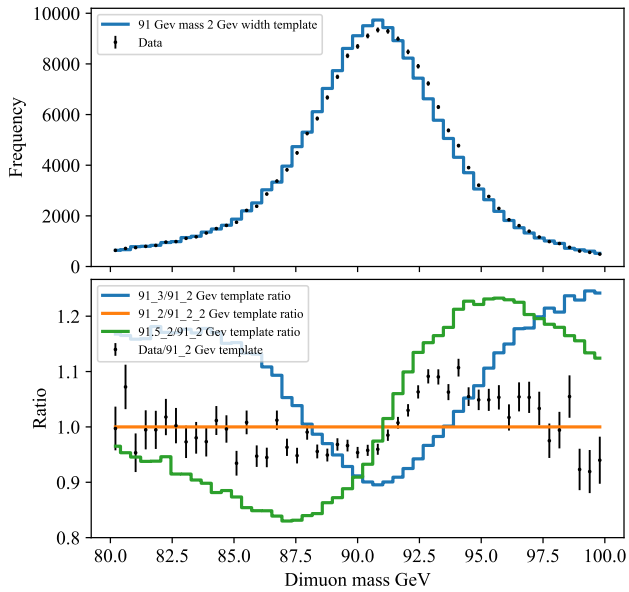

Figure 3:

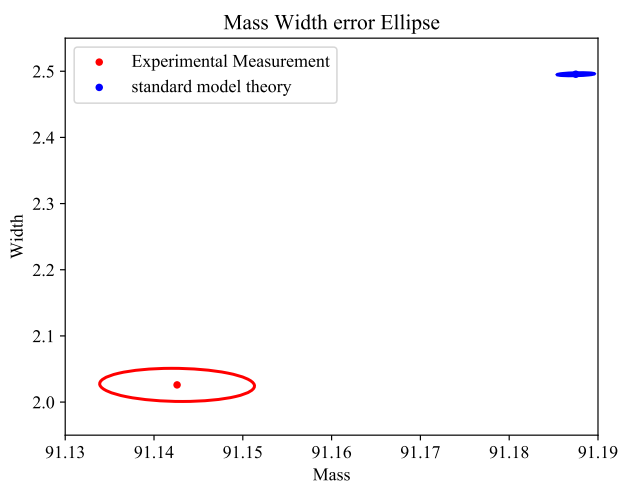

Figure 4:

When seperating the positve and negative dipole data:

Measuremnt	Positive dipole	Negative dipole	Significance
Mass [GeV]	91.10300 ± 0.01	91.16595 ± 0.01	3.49640
Width [GeV]	1.96170 ± 0.03	2.08124 ± 0.03	2.57377
χ^2	163.30	319.69	N/A
ndf	48	48	N/A

Table 4: Reults for positive and negative dipoles

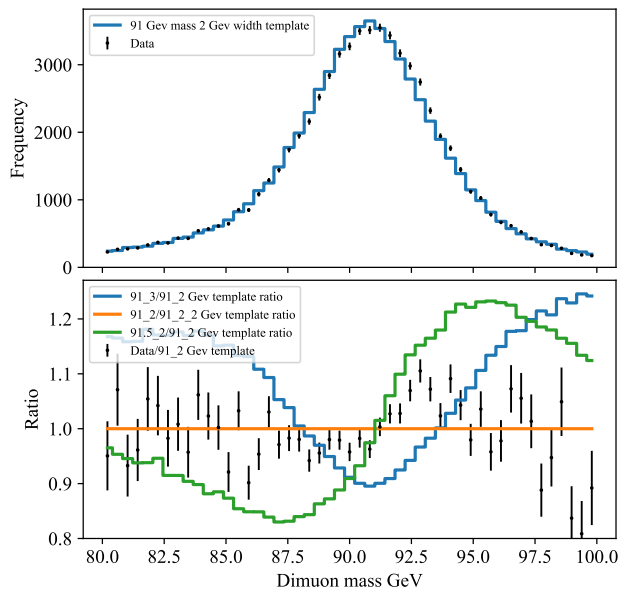

Figure 5: Positive dipole

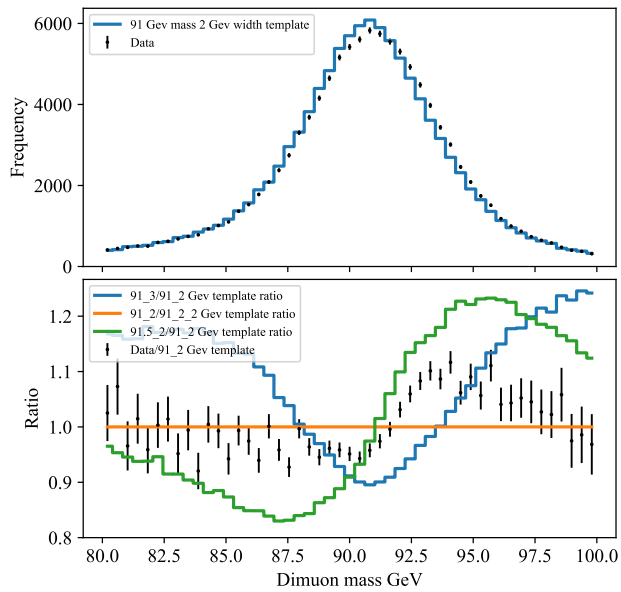

Figure 6: negative dipole

When seperating the 0-π and -π-0 angle data from the Positive muon:

Measuremnt	angle range 0-π	angle range-π-0	Significance
Mass [GeV]	91.02215 ± 0.01	91.26343 ± 0.01	13.85069
Width [GeV]	2.00701 ± 0.04	2.04523 ± 0.04	0.76370
χ^2	105.79	447.95	N/A
ndf	48	48	N/A

Table 5: Reults for angles between 0 and π and -π to 0

When seperating the Dimuon momenta between at the 18.3 Gev mark :

Measuremnt	0-18.3 Gev	$>$ 18.3Gev	Significance
Mass [GeV]	91.07786 ± 0.01	91.27105 ± 0.02	10.40806
Width [GeV]	1.97770 ± 0.03	2.17672 ± 0.04	3.95368
χ^2	187.67	282.93	N/A
ndf	48	48	N/A

Table 6: Reults for Dimuon transverse moemntum splitting up data at 18.3 Gev

When ajusting bin numbers to check for consistency

	Mass	Width
30 bins	91.27105 ± 0.01521	2.17672 ± 0.04338
50 bins	91.14260 ± 0.00872	2.02597 ± 0.02503
80 bins	91.14146 ± 0.00870	2.01189 ± 0.02496

Table 7: Z mass and width with statistical error for each bin size

selection cuts on the data Transverse momentum selection of ¿ 20 Gev per in both muons and pseudorapidity selection 2.2¡η ¡4.4

	Transverse momentum	pseudorapidity	Both applied simultaneously
mass	91.15046 ± 0.00876	91.13592 ± 0.01012	91.14296 ± 0.01016
width	2.00174 ± 0.02515	2.13729 ± 0.02930	2.11499 ± 0.02941
χ^2	424.649	339.732	348.053
χ^2	48.000	48.000	48.000
corelation	0.065	0.068	0.068

Table 8: Transver momentum selection of $>$ 20 Gev per in both muons and pseudorapidity selection 2.2$<\eta<$4.4